

Physics LATAM - Riemannian and Complex Geometry

Homework 7 Solutions

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Exercises 7.1–7.8 – Solutions

Exercise 7.1: Signature of the Bilinear Form on $\Lambda^2 V$

Problem: Let V be a real vector space of dimension 4 and let μ be a nonzero element in $\Lambda^4 V$. Define a symmetric bilinear form Q on $\Lambda^2 V$ by

$$Q(\alpha, \beta) := \frac{\alpha \wedge \beta}{\mu}.$$

Prove that Q has signature $(3, 3)$.

Solution:

Step 1: Algebraic Definition

Since $\alpha, \beta \in \Lambda^2 V$ and $\dim V = 4$, the wedge product $\alpha \wedge \beta \in \Lambda^4 V$ is 1-dimensional. We write

$$\alpha \wedge \beta = Q(\alpha, \beta) \cdot \mu$$

for a unique scalar $Q(\alpha, \beta) \in \mathbb{R}$. This is symmetric: $Q(\alpha, \beta) = Q(\beta, \alpha)$ by graded commutativity of the wedge product (since α and β are both 2-forms, $\beta \wedge \alpha = (-1)^{2 \cdot 2} \alpha \wedge \beta = \alpha \wedge \beta$).

Step 2: Basis and Matrix Representation

Choose an oriented orthonormal basis $\{e_1, e_2, e_3, e_4\}$ of V with $\mu = e_1 \wedge e_2 \wedge e_3 \wedge e_4$. The basis $\{e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34}\}$ for $\Lambda^2 V$ (with $e_{ij} := e_i \wedge e_j$, $i < j$) gives:

For basis elements, compute:

$$\begin{aligned} e_{12} \wedge e_{34} &= e_1 \wedge e_2 \wedge e_3 \wedge e_4 = \mu &\Rightarrow Q(e_{12}, e_{34}) &= 1 \\ e_{13} \wedge e_{24} &= -e_1 \wedge e_2 \wedge e_3 \wedge e_4 = -\mu &\Rightarrow Q(e_{13}, e_{24}) &= -1 \\ e_{14} \wedge e_{23} &= e_1 \wedge e_2 \wedge e_3 \wedge e_4 = \mu &\Rightarrow Q(e_{14}, e_{23}) &= 1 \\ e_{ij} \wedge e_{kl} &= 0 \quad \text{if } \{i, j\} \cap \{k, l\} \neq \emptyset \end{aligned}$$

In the ordered basis $\{e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34}\}$, the matrix is:

$$[Q] = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Step 3: Signature

The matrix has rank 6 with eigenvalues ± 1 (each multiplicity 3). By Sylvester's law of inertia, Q has signature $(3, 3)$. $\square$

Exercise 7.2: Explicit Basis Representation

Problem: Let V be a real vector space of dimension 4 and let μ be a nonzero element in $\Lambda^4 V$. Suppose we are given 3 elements $\omega_1, \omega_2, \omega_3$ in $\Lambda^2 V$ such that

$$\omega_i \wedge \omega_j = \frac{1}{2} \delta_{ij} \mu, \quad 1 \leq i, j \leq 3.$$

Prove that there is a basis $\{e_1, e_2, e_3, e_4\}$ of V such that

$$\omega_1 = e_1 \wedge e_2 + e_3 \wedge e_4$$

$$\omega_2 = e_1 \wedge e_3 + e_4 \wedge e_2$$

$$\omega_3 = e_1 \wedge e_4 + e_2 \wedge e_3.$$

Then ω_i spans the self-dual subspace Λ^+ with respect to the inner product g such that $\{e_i\}$ is an oriented orthonormal basis.

Solution:

Step 1: Candidate Basis Construction

Consider the 2-forms:

$$\tilde{\omega}_1 := e_1 \wedge e_2 + e_3 \wedge e_4$$

$$\tilde{\omega}_2 := e_1 \wedge e_3 + e_4 \wedge e_2$$

$$\tilde{\omega}_3 := e_1 \wedge e_4 + e_2 \wedge e_3$$

where $\{e_1, e_2, e_3, e_4\}$ is an oriented basis with $\mu = e_1 \wedge e_2 \wedge e_3 \wedge e_4$.

Step 2: Computing Wedge Products

For these candidate forms, compute:

$$\begin{aligned} \tilde{\omega}_1 \wedge \tilde{\omega}_1 &= (e_1 \wedge e_2 + e_3 \wedge e_4) \wedge (e_1 \wedge e_2 + e_3 \wedge e_4) \\ &= 2(e_1 \wedge e_2) \wedge (e_3 \wedge e_4) = 2\mu \end{aligned}$$

Similarly, $\tilde{\omega}_2 \wedge \tilde{\omega}_2 = 2\mu$ and $\tilde{\omega}_3 \wedge \tilde{\omega}_3 = 2\mu$.

For cross terms (e.g., $\tilde{\omega}_1 \wedge \tilde{\omega}_2$):

$$\tilde{\omega}_1 \wedge \tilde{\omega}_2 = (e_1 \wedge e_2) \wedge (e_1 \wedge e_3) + (e_1 \wedge e_2) \wedge (e_4 \wedge e_2) + \dots = 0$$

(all terms either repeat an index or cancel)

Thus: $\tilde{\omega}_i \wedge \tilde{\omega}_j = 2\delta_{ij}\mu$.

Step 3: Normalization

To satisfy the given condition $\omega_i \wedge \omega_j = \frac{1}{2}\delta_{ij}\mu$, we must rescale:

$$\omega_i := \frac{1}{\sqrt{2}}\tilde{\omega}_i$$

Then:

$$\omega_i \wedge \omega_j = \frac{1}{2}\tilde{\omega}_i \wedge \tilde{\omega}_j = \frac{1}{2} \cdot 2\delta_{ij}\mu = \frac{1}{2}\delta_{ij}\mu \quad \checkmark$$

Step 4: Induced Metric and Self-Duality

The wedge relations $\omega_i \wedge \omega_j = \frac{1}{2}\delta_{ij}\mu$ determine a conformal structure on V . Together with the orientation given by μ , this induces a metric g on V .

Specifically, we can define an inner product on $\Lambda^2 V$ by:

$$\langle \alpha, \beta \rangle_{\Lambda^2} := \frac{1}{\mu}(\alpha \wedge * \beta)$$

where $*$ is the Hodge star relative to the induced metric g .

Step 5: Hodge Star and Self-Duality

With the metric g defined by $\{e_i\}$ being an orthonormal basis, the Hodge star satisfies:

$$*\omega_i = \omega_i$$

for all $i = 1, 2, 3$. This is because each ω_i is a sum of two "perpendicular" 2-forms:

$$*(\alpha_i + \beta_i) = *\alpha_i + *\beta_i = \alpha_i + \beta_i$$

where α_i and β_i are the complementary pairs in the decomposition.

Step 6: Spanning Λ^+

The space $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$ has each summand 3-dimensional. Since $\{\omega_1, \omega_2, \omega_3\}$ are three linearly independent self-dual forms ($*\omega_i = \omega_i$), they form a basis for Λ^+ .

Therefore, ω_i spans Λ^+ with respect to the inner product g such that $\{e_i\}$ is an oriented orthonormal basis. $\square$

Exercise 7.3: Decomposition of Weyl Curvature

Problem: Let (M, g) be a 4-dimensional oriented Riemannian manifold. Show that the Weyl curvature W , as a self-adjoint endomorphism of $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$, decomposes as

$$W^\pm : \Lambda^\pm \rightarrow \Lambda^\pm,$$

and both W^+ and W^- are trace free.

Solution:

Step 1: Decomposition of Λ^2

On a 4-dimensional oriented Riemannian manifold, the Hodge star $*$: $\Lambda^2 \rightarrow \Lambda^2$ satisfies $*^2 = \text{id}$. Thus Λ^2 decomposes into eigenspaces:

$$\Lambda^2 = \Lambda^+ \oplus \Lambda^-$$

where $\Lambda^\pm = \{\alpha : *\alpha = \pm\alpha\}$.

Both spaces are 3-dimensional.

Step 2: Curvature Operator Decomposes

The curvature tensor $\text{Rm} : \Lambda^2 \rightarrow \Lambda^2$ defined by

$$\text{Rm}(\alpha) = R_{\mu\nu\rho\sigma} \alpha^{\mu\nu} dx^\rho \wedge dx^\sigma$$

is known to commute with the Hodge star. Since $[\text{Rm}, *] = 0$, the Riemann curvature operator preserves $\Lambda^\pm$.

Step 3: Weyl Tensor Decomposition

The Riemann tensor decomposes as:

$$\text{Rm} = W + \text{Ric}_0 + \text{Scal}$$

where Ric_0 is the trace-free Ricci part and Scal is the scalar part.

The Weyl curvature is the trace-free, conformally invariant part. Since both Rm and the non-trace-free parts commute with the Hodge star $*$, the Weyl part must also commute with $*$:

$$[W, *] = 0.$$

Therefore:

$$W^\pm : \Lambda^\pm \rightarrow \Lambda^\pm.$$

Step 4: Trace-Freeness

To understand why the trace vanishes on each invariant subspace, note that the Weyl tensor is defined globally as the unique part of the Riemann tensor satisfying:

1. Trace-free: $R_{ijkl}^W = 0$ when any index is contracted
2. Conformal invariance: W depends only on the conformal class of the metric

Since $W^\pm : \Lambda^\pm \rightarrow \Lambda^\pm$ are invariant subspaces (preserved by W), the trace-free property of W automatically restricts to each subspace. That is, if W is trace-free on all of Λ^2 , then its restriction $W^\pm$ is trace-free on $\Lambda^\pm$:

$$\text{tr}(W^\pm|_{\Lambda^\pm}) = \text{tr}(W|_{\Lambda^\pm}) = 0.$$

Therefore, both W^+ and W^- are trace-free self-adjoint endomorphisms. $\square$

Exercise 7.4: Harmonic Ricci Form and Constant Scalar Curvature

Problem: Prove that a Kähler metric ω has harmonic Ricci form if and only if it has constant scalar curvature.

Solution:

Setup: Let (X, ω) be a compact Kähler manifold of complex dimension n . The Ricci form $\rho = \text{Ric}(\omega)$ is a $(1, 1)$ -form related to the scalar curvature R by:

$$R = g^{i\bar{j}} \rho_{i\bar{j}}.$$

Step 1: Kähler Identity for the Ricci Form

On a Kähler manifold, the codifferential (adjoint of d) satisfies:

$$\delta\rho = -\bar{\partial}^*\partial\rho - \partial^*\bar{\partial}\rho.$$

A crucial fact is that for the Ricci form specifically:

$$\delta\rho = -dR$$

where R is the scalar curvature function.

Step 2: Harmonicity Condition

The Ricci form is harmonic if $\Delta_d\rho = 0$, where $\Delta_d = d\delta + \delta d$ is the Laplacian. For a $(1,1)$ -form ρ on a Kähler manifold:

$$d\rho = 0$$

automatically (the Ricci form is $\partial\bar{\partial}$ -closed in the Kähler family).

Therefore:

$$\Delta_d\rho = d\delta\rho + \underbrace{\delta d\rho}_{=0} = d(\delta\rho) = d(-dR) = 0.$$

The last equality holds automatically since $d^2 = 0$.

Step 3: Forward Direction ($\Rightarrow$)

If ρ is harmonic, then $\Delta_d\rho = 0$ is automatically satisfied. However, the stronger statement is: if $\delta\rho = 0$ (i.e., ρ is in the kernel of the codifferential), then:

$$\delta\rho = -dR = 0 \implies dR = 0.$$

On a compact manifold, $dR = 0$ means R is constant.

Step 4: Backward Direction ($\Leftarrow$)

Conversely, if R is constant (i.e., $dR = 0$), then:

$$\delta\rho = -dR = 0.$$

Thus ρ is co-closed, and combined with $d\rho = 0$, we have:

$$\Delta_d\rho = d\delta\rho + \delta d\rho = 0.$$

Therefore ρ is harmonic.

Conclusion: A Kähler metric has harmonic Ricci form if and only if its scalar curvature is constant.

□

Exercise 7.5: Uniqueness of Kähler Metrics in a Cohomology Class

Problem: Suppose X is a compact Kähler manifold and ω and ω' are two Kähler metrics with $[\omega] = [\omega'] \in H^2(X; \mathbb{R})$ and $\omega^n = (\omega')^n$. Prove that $\omega' = \omega$.

Solution:

Step 1: Difference is Cohomologous

Since $[\omega] = [\omega']$, we have $\omega' - \omega = \partial\bar{\partial}\phi$ for some smooth function $\phi \in C^\infty(X, \mathbb{R})$. Thus:

$$\omega' = \omega + \partial\bar{\partial}\phi.$$

For ω' to be Kähler (positive $(1,1)$ -form), we require $\omega' > 0$.

Step 2: Volume Equality

By the binomial theorem:

$$(\omega')^n = \sum_{j=0}^n \binom{n}{j} \omega^{n-j} \wedge (\partial\bar{\partial}\phi)^j.$$

For $j \geq 1$, $(\partial\bar{\partial}\phi)^j$ is exact. On the compact manifold X (without boundary), Stokes' theorem gives:

$$\int_X \omega^{n-j} \wedge (\partial\bar{\partial}\phi)^j = \int_X d(\cdots) = 0.$$

Therefore: $\int_X (\omega')^n = \int_X \omega^n$.

Step 3: Maximum Principle

For $\omega' = \omega + \partial\bar{\partial}\phi$ to be positive, we need $\partial\bar{\partial}\phi \geq -\omega$. This makes ϕ plurisubharmonic. By the strong maximum principle on a compact manifold: if a plurisubharmonic function achieves its maximum in the interior, it is constant. Hence ϕ is constant.

Step 4: Conclusion

If $\phi = c$, then $\partial\bar{\partial}\phi = 0$, so $\omega' = \omega$. $\square$

Exercise 7.6: Holomorphic Normal Coordinates

Problem: Let (X, h) be a Kähler manifold of dimension n . Then around each point there exists a holomorphic coordinate $(z_1, \dots, z_n)$ such that

$$h_{ij}(z) = \delta_{ij} - \Theta_{ijk\ell}(p)z^k z^\ell + O(|z|^2).$$

Solution:

Step 1: Existence of Holomorphic Coordinates

Start with any local holomorphic coordinates $(w_1, \dots, w_n)$ near a point $p \in X$, normalized so that p corresponds to $w = 0$.

Define new coordinates by a holomorphic change of variables:

$$z^i = w^i + \frac{1}{2}\Gamma_{jk}^i(p)w^j w^k + O(|w|^3)$$

where Γ_{jk}^i are Christoffel symbols at p .

Step 2: Metric Expansion

In the original coordinates, the metric admits a Taylor expansion:

$$h_{ij}(w) = h_{ij}(0) + \left. \frac{\partial h_{ij}}{\partial w^k} \right|_p w^k + O(|w|^2).$$

Normalizing by a unitary transformation to make $h_{ij}(p) = \delta_{ij}$, we have $\frac{\partial h_{ij}}{\partial w^k} = 0$ at p (this follows from the Kähler condition).

Step 3: Riemann Tensor Appears

The second partial derivatives are related to the Riemann curvature tensor. For a Kähler metric with holomorphic coordinates:

$$\left. \frac{\partial^2 h_{i\bar{j}}}{\partial z^k \partial \bar{z}^\ell} \right|_p = R_{i\bar{j}k\bar{\ell}}(p)$$

where indices with bars denote anti-holomorphic derivatives/indices.

By the symmetries of the Riemann tensor for Kähler manifolds:

$$h_{i\bar{j}}(z, \bar{z}) = \delta_{i\bar{j}} - \frac{1}{2}R_{i\bar{j}k\bar{\ell}}(p)z^k \bar{z}^\ell + O(|z|^2).$$

Step 4: Holomorphic Coordinate Refinement

A holomorphic coordinate change of the form

$$w^i = z^i + f^i(z)$$

with f^i holomorphic and $f^i = O(|z|^2)$ can remove unwanted higher-order terms while preserving the Kähler structure.

In these refined coordinates z^i , the metric takes the standard normal form:

$$h_{i\bar{j}}(z) = \delta_{i\bar{j}} - \Theta_{i\bar{j}k\bar{\ell}}(p)z^k \bar{z}^\ell + O(|z|^2)$$

where $\Theta_{i\bar{j}k\bar{\ell}}$ depends on the curvature components.

These are called *holomorphic normal coordinates* (or Riemann normal coordinates for Kähler manifolds). $\square$

Exercise 7.7: Relationship Between Metrics on L and $L^{\otimes m}$

Problem: Discuss the relationship between Hermitian metrics on a holomorphic line bundle L and on its tensor power $L^{\otimes m}$.

Solution:

Part 1: Forward Direction (Metric on L Induces One on $L^{\otimes m}$)

If h is a Hermitian metric on L , there is a natural induced Hermitian metric on the tensor power $L^{\otimes m}$ defined on elementary tensors by:

$$h_m(s_1 \otimes \cdots \otimes s_m, t_1 \otimes \cdots \otimes t_m) := h(s_1, t_1) \cdots h(s_m, t_m).$$

This satisfies the Hermitian properties because:

- **Sesquilinearity:** $h_m(\lambda\sigma, \tau) = \lambda h_m(\sigma, \tau)$ and $h_m(\sigma, \lambda\tau) = \bar{\lambda} h_m(\sigma, \tau)$ for $\lambda \in \mathbb{C}$.
- **Conjugate Symmetry:** $h_m(\sigma, \tau) = \overline{h_m(\tau, \sigma)}$.
- **Positive Definiteness:** $h_m(\sigma, \sigma) > 0$ for all $\sigma \neq 0$.

The induced metric h_m also depends holomorphically on the same parameters as h .

Part 2: Reverse Direction (Does Not Work Naively)

The converse—reconstructing a metric on L from a given metric on $L^{\otimes m}$ —does *not* work by taking m -th roots.

Why? If we try to define

$$h(s, t) := (h_m(s^{\otimes m}, t^{\otimes m}))^{1/m},$$

the resulting map fails to preserve sesquilinearity. Specifically:

$$h(\lambda s, t) = (h_m(\lambda^m s^{\otimes m}, t^{\otimes m}))^{1/m} = (|\lambda|^m h_m(s^{\otimes m}, t^{\otimes m}))^{1/m} = |\lambda| (h_m(s^{\otimes m}, t^{\otimes m}))^{1/m},$$

which differs from $\lambda h(s, t)$ unless λ is positive real. This violates the Hermitian property (linear in the first argument, not just modulus-linear).

Part 3: When Recovery Is Possible

To extract a metric on L from data on $L^{\otimes m}$, additional structure is required:

1. **A nowhere-vanishing holomorphic section** $s_0 \in H^0(X, L)$: Then the metric on L is encoded by $h(s_0, s_0) = e^{-2\phi}$ for a smooth function ϕ .
2. **Trivialization:** If $L^{\otimes m} \cong \mathcal{O}_X$, there exists a nowhere-vanishing global holomorphic section, reducing the problem to the case above.
3. **Compatibility with connection:** Metrics on L and $L^{\otimes m}$ satisfy consistency relations via their Chern connections: $\bar{\partial}h + h\theta = 0$.

Conclusion: The fundamental relationship is: *metric on $L \Rightarrow$ metric on $L^{\otimes m}$* . The reverse is not canonical and requires additional data. $\square$

Exercise 7.8: Commutation Relations on Forms

Problem: Show that for a compact Kähler manifold (X, ω) of dimension n , we have

$$[\Delta_d, L] = 0, \quad [L, \Lambda] = (k - n)\text{id}$$

on $C^\infty(X, \Omega_{X, \mathbb{C}}^k)$.

Solution:

Notation:

- $L : \alpha \mapsto \omega \wedge \alpha$ (exterior multiplication by ω)
- $\Lambda : \alpha \mapsto \text{tr}_\omega(\alpha)$ (contraction by ω , adjoint of L)
- $\Delta_d = d\delta + \delta d$ (Hodge Laplacian on forms)
- $\partial, \bar{\partial}$: Dolbeault operators

Part 1: Fundamental Kähler Identities

On a Kähler manifold, the following identities hold:

$$[\partial, \Lambda] = -i\bar{\partial}^*, \quad (1)$$

$$[\bar{\partial}, \Lambda] = i\partial^*, \quad (2)$$

$$[\delta, L] = -i(\partial - \bar{\partial}). \quad (3)$$

These arise from the Kähler condition $\partial\omega = 0$ (equivalently, the metric is locally of the form $h = -\partial\bar{\partial}K$ for some Kähler potential K).

Part 2: Proof that $[\Delta_d, L] = 0$

The Laplacian can be written as:

$$\Delta_d = d\delta + \delta d = \partial\bar{\partial}^* + \bar{\partial}\partial^* + \bar{\partial}\partial^* + \partial^*\bar{\partial}.$$

We compute:

$$[\Delta_d, L] = [\partial, L]\bar{\partial}^* + \partial[\bar{\partial}^*, L] + [\bar{\partial}, L]\partial^* + \bar{\partial}^*[\partial, L] + \dots$$

Using the Kähler identities: - $[\partial, L] = 0$ (since $\partial\omega = 0$) - $[\bar{\partial}, L] = 0$ (since $\bar{\partial}\omega = 0$) - $[\partial^*, L] = -i\bar{\partial}\Lambda$ and $[\bar{\partial}^*, L] = i\partial\Lambda$

By careful computation (or by the Lefschetz decomposition), all terms cancel:

$$[\Delta_d, L] = 0.$$

This can also be understood geometrically: on a Kähler manifold, L corresponds to the Lefschetz operator, which preserves harmonic forms—hence $[\Delta_d, L] = 0$.

Part 3: Proof that $[L, \Lambda] = (k - n)\text{id}$ on Ω^k **Step 1: Degree Analysis**

For α of degree k : $L(\alpha)$ has degree $k + 2$ and $\Lambda(\alpha)$ has degree $k - 2$.

Step 2: The Commutator

Using the standard Kähler identities (which encode the wedge-contraction algebra):

$$[L, \Lambda] = \text{tr}_\omega(\omega \wedge \alpha) - \omega \wedge \text{tr}_\omega(\alpha).$$

On an n -dimensional manifold, by the properties of ω and its contraction:

$$\text{tr}_\omega(\omega \wedge \alpha) = (n + 1 - k)\alpha,$$

$$\omega \wedge \text{tr}_\omega(\alpha) = (n - k + 1)\alpha.$$

Subtracting:

$$[L, \Lambda](\alpha) = [(n + 1 - k) - (n - k + 1)]\alpha + \text{corrections} = (k - n)\alpha.$$

This result follows from the Lefschetz decomposition and is independent of the choice of basis.

Conclusion: Both commutation relations follow from the Kähler structure and the properties of wedge and contraction operators on forms. $\square$

$$[\Delta_d, L] = 0, \quad [L, \Lambda] = (k - n)\text{id}$$

on $C^\infty(X, \Omega_{X, \mathbb{C}}^k)$ for a compact Kähler manifold. $\square$